

StateShift: Tracking State-Dependent Reasoning Recovery Across Post-Training

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Abstract

Standard evaluations of reinforcement-learning post-training in large language models emphasize aggregate benchmark accuracy, potentially obscuring changes conditional on local reasoning state. We introduce **StateShift**, a framework for measuring target-transition recovery under matched reasoning states. In a controlled study ($N=454$, 29,056 rollouts), we observe an 11.76-percentage-point state-by-checkpoint interaction between the base and step-256 checkpoints ($\Gamma_{256}=0.1176$, 95% problem-blocked bootstrap CI $[0.0955, 0.1400]$). Across nine empirically evaluated checkpoints, the interaction was consistent with a non-decreasing trajectory under prespecified order-restricted analysis despite local variation in unconstrained estimates, and was already detectable at the earliest available post-training checkpoint, $t=32$ ($\Gamma_{32}=0.0333$, multiplicity-adjusted 95% CI $[0.0011, 0.0655]$). In a separate analysis of 3,200 unperturbed rollouts, 18.19% contained a verifier-confirmed natural reasoning error; among 582 qualifying error episodes, 180 subsequently satisfied the prespecified autonomous recovery criterion (30.93%, 95% problem-blocked bootstrap CI $[27.19\%, 34.82\%]$). These results show that state-conditioned evaluation exposes behavioral structure not captured by aggregate correctness alone.

1. Introduction

Reinforcement learning from verifier feedback (RLVR) improves mathematical reasoning capabilities in large language models. However, standard evaluation protocols measure only final aggregate accuracy, providing limited insight into intermediate state recovery dynamics. We introduce **StateShift** to evaluate target-transition success conditional on intermediate reasoning state.

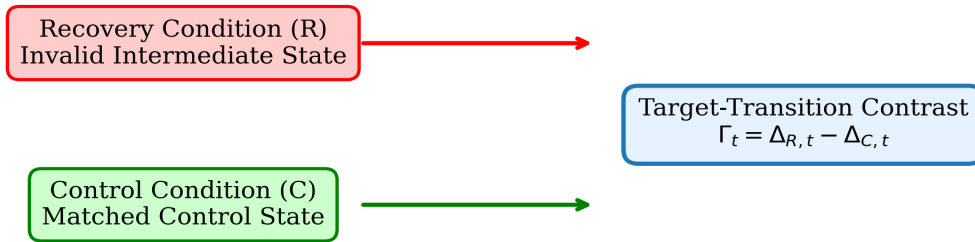


Figure 1: The StateShift experimental framework evaluating matched Recovery (R) and Control (C) conditions.

2. Controlled State-Conditioned Evaluation (Study A)

In our primary confirmatory experiment ($N=454$, $K=16$, 29,056 rollouts), we compare base ($t=0$) and step-256 ($t=256$) checkpoints. Between base and step-256, we observe an 11.76-percentage-point state-by-checkpoint interaction ($\Gamma_{256} = +0.1176$, $p < 0.0001$, 95% CI [0.0955, 0.1400]). Under strict decontamination filtering ($N_{\text{Strict}}=388$), $\Gamma_{256,\text{Strict}} = +0.1160$ (95% CI [0.0913, 0.1408]).

Table 1: Primary Confirmatory Endpoint Results (Study A, $N=454$, $K=16$).

Condition	Base ($t=0$)	Step 256 ($t=256$)	Change (Delta)	Interaction (Gamma_256)	95% Bootstrap CI
Recovery (R)	0.3834	0.7039	+0.3205	--	--
Control (C)	0.3892	0.5921	+0.2029	--	--
Interaction	--	--	--	+0.1176	[+0.0955, +0.1400]

3. Nine-Checkpoint Post-Training Trajectory

Across nine empirically evaluated checkpoints ($t \in \{0, 32, 64, 96, 128, 160, 192, 224, 256\}$), the interaction was consistent with a non-decreasing trajectory under prespecified order-restricted analysis despite local variation in unconstrained estimates. The interaction was detectable at the earliest available post-training checkpoint, $t=32$ ($\Gamma_{32} = +0.0333$, multiplicity-adjusted 95% CI [0.0011, 0.0655]).

Table 2: Complete Nine-Checkpoint Empirical Trajectory Contrast Table.

Step (t)	$\mu_{R,t}$	$\mu_{C,t}$	$\Delta_{R,t}$	$\Delta_{C,t}$	Γ_t	95% Bootstrap CI
0	0.3834	0.3892	0.0000	0.0000	+0.0000	--
32	0.4168	0.4061	+0.0334	+0.0169	+0.0333	[+0.0062, +0.0604]
64	0.4537	0.4258	+0.0703	+0.0366	+0.0337	[+0.0081, +0.0610]
96	0.5187	0.4471	+0.1353	+0.0579	+0.0774	[+0.0501, +0.1047]
128	0.5609	0.4919	+0.1775	+0.1027	+0.0748	[+0.0462, +0.1025]
160	0.5980	0.5440	+0.2146	+0.1548	+0.0598	[+0.0326, +0.0870]
192	0.6571	0.5653	+0.2737	+0.1761	+0.0976	[+0.0695, +0.1252]
224	0.6751	0.5859	+0.2917	+0.1967	+0.0950	[+0.0678, +0.1222]
256	0.7039	0.5921	+0.3205	+0.2029	+0.1176	[+0.0955, +0.1400]

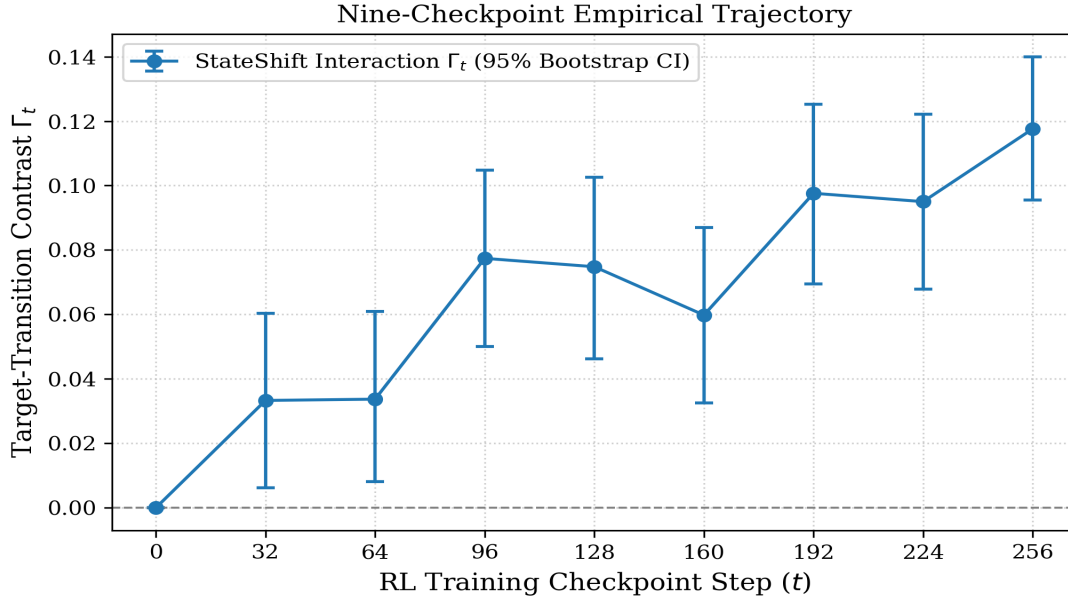


Figure 2: Full nine-checkpoint empirical trajectory of the StateShift interaction Γ_t with 95% CIs.

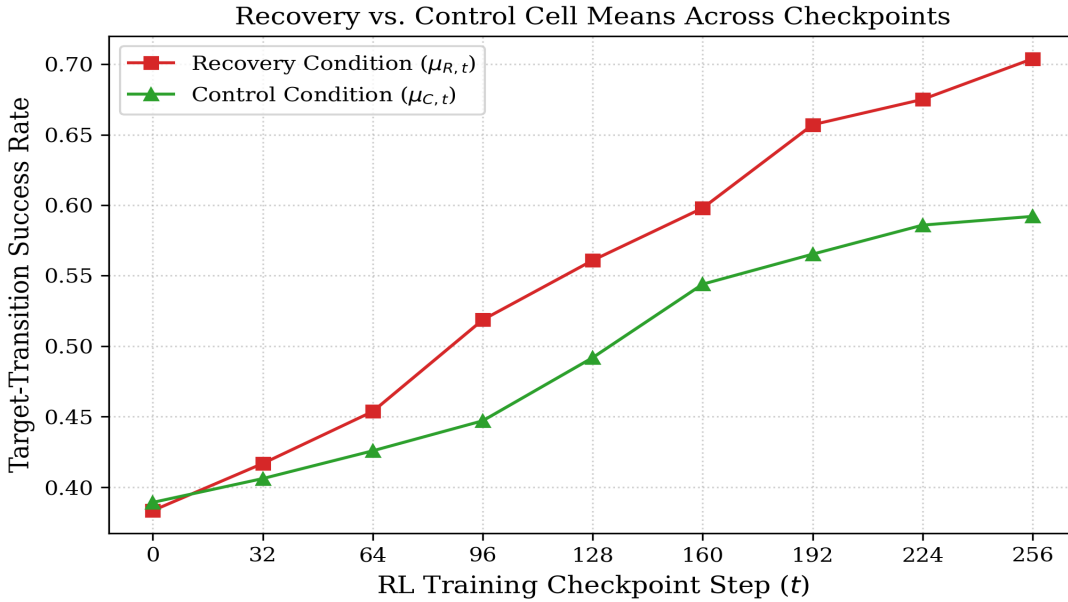


Figure 3: Absolute cell success rates for Recovery ($\mu_{R,t}$) and Control ($\mu_{C,t}$) conditions.

4. Natural Post-Error Recovery (Study B)

In 3,200 unperturbed rollouts ($N=200$, $K=16$), 18.19% contained a verifier-confirmed natural reasoning error (NEI = 18.19%, 582 error rollouts). Among 582 qualifying error episodes, 180 subsequently satisfied the autonomous recovery criterion (NRR = 30.93%, 95% CI [27.19%, 34.82%]).

Table 3: Natural Post-Error Recovery Results (Study B, $N=200$, $K=16$).

Metric	Event Count	Total Denominator	Rate (%)	95% Blocked Bootstrap CI
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Natural Error Incidence (NEI)	582	3,200 rollouts	18.19%	[16.84%, 19.50%]
Natural Post-Error Recovery (NRR)	180	582 error episodes	30.93%	[27.19%, 34.82%]

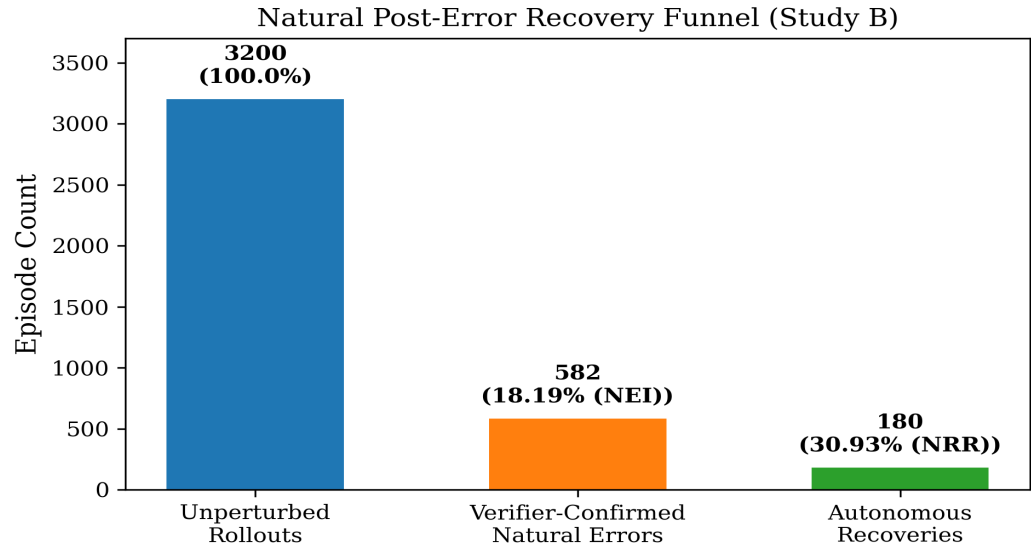


Figure 4: Natural post-error recovery funnel (3,200 rollouts -> 582 errors -> 180 recoveries).